

Extreme Precipitation Risk Modelling Using Generalized Pareto Distribution (GPD)

Technical Documentation of R Implementation

This document describes an R script used to perform Extreme Value Analysis (EVA) on precipitation data derived from climate projections and historical observations. The model uses a Peaks Over Threshold (POT) approach combined with a Generalized Pareto Distribution (GPD) to estimate extreme precipitation return levels under multiple climate scenarios. The workflow integrates declustering, stationarity testing, and automated model selection to quantify extreme rainfall risk across historical and climate projection scenarios. The script is designed to support flood-risk and climate impact analysis by estimating return levels for long recurrence intervals such as 10, 100, and 1000-year precipitation events.

Dataset Description

Input datasets consist of daily precipitation time series derived from previously processed NetCDF climate projection data. Scenarios analyzed include: • Historical climate observations (1971–2024) • RCP 2.6 climate projection (2025–2080) • RCP 4.5 climate projection (2025–2080) • RCP 8.5 climate projection (2025–2080) Each dataset contains two main variables: • Date – observation date • Precipitation_mm_day – spatially averaged daily precipitation (mm/day) These datasets are produced by a separate ETL processing pipeline which extracts precipitation values from Copernicus climate projection NetCDF files and aggregates them into CSV time series.

Extreme Value Modelling Pipeline

The modelling workflow follows a structured Extreme Value Analysis pipeline: 1. Data ingestion 2. Date preprocessing and validation 3. Event declustering 4. Threshold exceedance selection 5. GPD model estimation 6. Stationarity testing 7. Return level estimation 8. Scenario comparison and results storage

1. Library Dependencies

The script relies on the following R packages: • POT – declustering and Peaks Over Threshold tools • extRemes – extreme value modelling and GPD estimation • data.table – efficient tabular data manipulation These packages provide the statistical infrastructure required to perform extreme value modelling and large dataset handling.

2. Scenario Definition and Data Loading

Four precipitation datasets are loaded corresponding to historical observations and three future climate scenarios. Each scenario is stored inside a structured data.table containing: • Scenario name • CSV file source • Dataset • Time period • Predefined modelling thresholds This structure allows automated iteration across multiple scenarios using a single modelling loop.

3. Date Processing and Data Validation

The Date column of each dataset is converted from character format to Date objects. The script includes checks for missing values introduced during conversion. Any rows with invalid dates are removed before modelling. The duration of each dataset is calculated in years using the difference between minimum and maximum observation dates.

4. Declustering of Extreme Events

Extreme precipitation events often occur in multi-day storms. Because Extreme Value Theory assumes independent observations, declustering is required to remove temporal dependence. The `POT::clust()` function is used with:

- threshold = 90th percentile of wet-day precipitation
- time window = 3 days

This groups exceedances occurring within three days into clusters representing individual storm events.

5. Cumulative Event Magnitude Calculation

Instead of using daily maxima, the script computes cumulative precipitation within each storm cluster. For each cluster:

- All precipitation values are summed
- The timestamp of the peak rainfall event is recorded

This produces a dataset of independent storm events characterized by their total precipitation magnitude.

6. Threshold Selection for GPD

Extreme value modelling requires selecting a threshold above which observations are treated as extremes. Two threshold approaches are supported:

- Anchored threshold (fixed across scenarios)
- Diagnostic threshold (derived from exploratory plots)

The threshold determines which cluster precipitation events are used in the Peaks Over Threshold model.

7. Generalized Pareto Distribution Modelling

The Generalized Pareto Distribution (GPD) is fitted to precipitation exceedances using the `extRemes::fevd()` function. Two models are estimated:

- Stationary model:
 - Assumes distribution parameters remain constant over time
- Non-stationary model:
 - Allows the scale parameter to vary as a function of time

Time is standardized to avoid numerical instability during estimation.

8. Stationarity Testing and Model Selection

A likelihood ratio test compares stationary and non-stationary models. Decision rule:

- $p\text{-value} < 0.05 \rightarrow$ non-stationary model selected
- $p\text{-value} \geq 0.05 \rightarrow$ stationary model retained

This ensures the final model reflects potential climate-driven changes in extreme precipitation behavior.

9. Return Level Estimation

Return levels represent precipitation magnitudes expected to be exceeded once every N years. Return levels are estimated for:

- 10-year events
- 20-year events
- 50-year events
- 100-year events
- 250-year events
- 500-year events
- 1000-year events

For non-stationary models, return levels are evaluated at the end of the scenario time period to represent future risk.

10. Confidence Intervals

Return level estimates include 95% confidence intervals calculated using the `ci()` function. Each result contains:

- Return level estimate
- Lower confidence bound
- Upper confidence bound

These intervals capture statistical uncertainty in extreme value estimation.

11. Results Storage Structure

Model outputs are stored in structured lists and data tables. For each scenario the script stores:

- Selected threshold
- Model type (stationary or non-stationary)
- Fitted GPD model object
- Return level estimates
- Confidence intervals

This structure enables later visualization, scenario comparison, and integration with downstream flood risk models.

12. Scenario Risk Comparison

The final section extracts selected return levels for comparison between scenarios. Examples include: • 10-year precipitation events • 100-year precipitation events • 1000-year extreme events Comparing these metrics across scenarios allows assessment of how extreme rainfall risk may evolve under different climate pathways.

13. Application to Flood Risk Modelling

The estimated return levels can be integrated into flood risk frameworks. Applications include: • urban flood hazard modelling • infrastructure resilience assessment • agricultural damage analysis • climate stress testing for financial portfolios In spatial modelling workflows, these return levels can be combined with exposure and vulnerability layers within GIS systems to compute climate risk metrics.